Fine-tuning SIMCLR using a medical dataset

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January 5, 2025

Introduction

- SimCLR is a contrastive self-supervised learning framework introduced by Chen et al. (2020)
- Relies heavily on strong data augmentations, particularly color-based augmentations such as color jittering and random color distortions, to create positive pairs.
- A crucial aspect of SimCLR is that color augmentations are not optional, the original paper explicitly shows that removing color distortions leads to a large drop in performance on natural image benchmarks.

Motivation

- If color is a primary signal used during representation learning, will SimCLR generalize to domains where color is absent or meaningless?
- Medical images such as chest X-rays differ significantly from natural images:
  - They are typically grayscale or near-grayscale
  - Image transformations that change or destroy those medical signals may distort clinically meaningful information

Objective

To observe

- How well SimCLR representations pre-trained on non-medical images (CIFAR-100) transfer to medical imaging tasks.
- The performance difference between linear evaluation and fine-tuning when labeled data is limited.
- Whether domain similarity (medical vs. non-medical) affects downstream performance when data size is controlled.

Objective

- To answer these questions, three controlled experiments were conducted:
  - Experiment 1: Fine-tuning of a SimCLR-pretrained model on a medical dataset.
  - Experiment 2: Linear evaluation of a SimCLR-pretrained model on the same medical dataset.
  - Experiment 3: Linear evaluation of a SimCLR-pretrained model on a large non-medical dataset, using matched data splits for fair comparison.

All experiments use the same backbone architecture and pretrained weights to ensure consistency.

Dataset Descriptions

Medical Dataset: Chest X-Ray Pneumonia

- Source: Kaggle Chest X-ray Pneumonia Dataset
- Domain: Medical imaging (grayscale X-rays)
- Classes: NORMAL, PNEUMONIA
- Total samples used: 3306

Data Splits

- Training: 2413 downsized from 4694 (1206- Pneumonia and 1207- Normal)
- Validation: 269
- Test: 624

Class-weighted cross-entropy loss addressed imbalance.

Dataset Descriptions

Non-Medical Dataset: Cats vs Dogs

- Source: Kaggle Cats vs Dogs Dataset
- Domain: Natural, RGB images
- Classes: Cat, Dog
- Total samples used: 5,840 (Downsized from 25k)
- To ensure fair comparison with the medical dataset, identical split sizes were enforced.

Data Splits:

- Training: 4,694
- Validation: 522
- Test: 624

Training Setup

The pretrained model used in this project:

- Architecture: ResNet-50
- Pre-training dataset: CIFAR-100
- edadaltocg/resnet50_simclr_cifar100

Training Hyperparameters

- Optimizer: Adam
- Learning Rate:0.001
- Epochs: 6
- Batch Size: 64

Experiment 1: Fine-Tuning on Medical Data

- Evaluate whether allowing limited model adaptation improves performance on a medical task with domain shift.
- Freezing most of the weights.
- Unfreezing only the last residual block.
- Training both the classifier and a small portion of the model.
- Class-weighted loss used

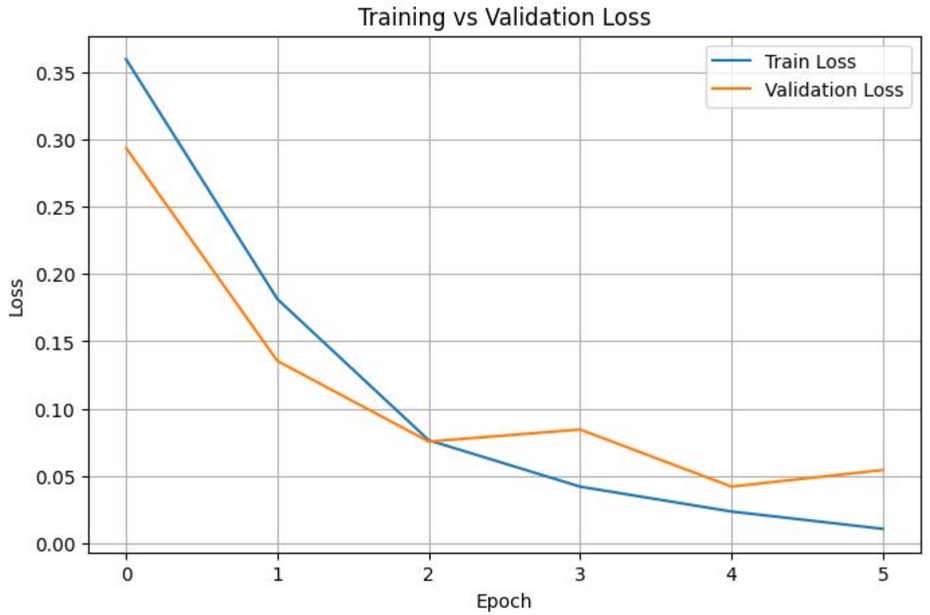
Experiment 1: Fine-Tuning on Medical Data

Final Test Performance

- Test Accuracy: 84.29%

Classification Report Highlights

- PNEUMONIA:
  - Recall: 0.99 (excellent sensitivity)
- NORMAL:
  - Recall: 0.58 (lower, indicating bias toward pneumonia detection)



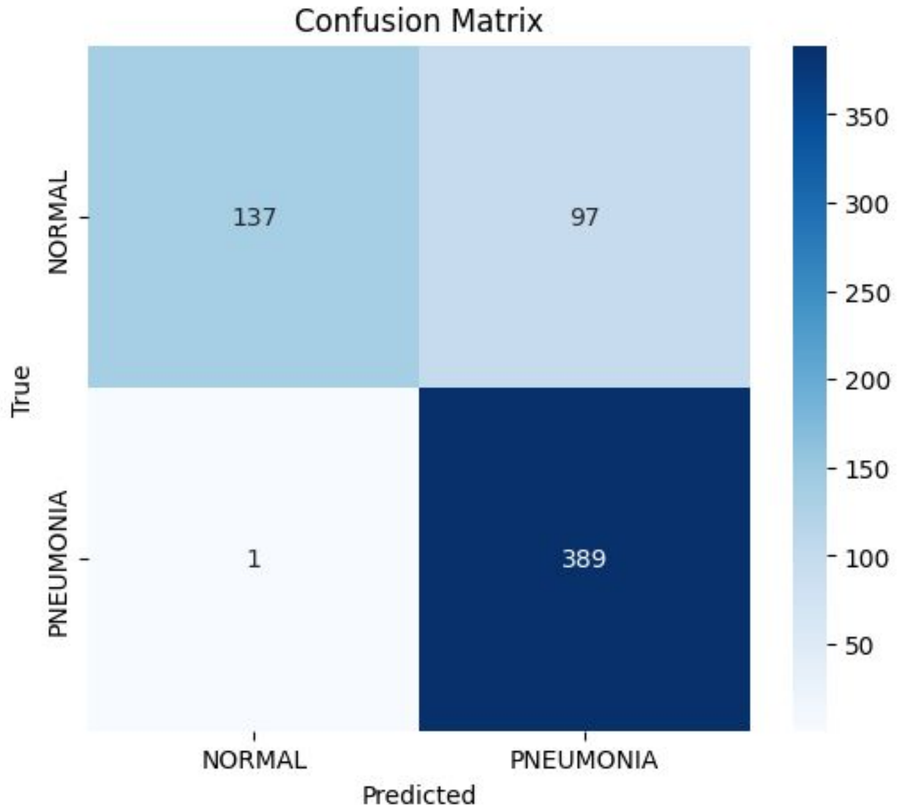
Experiment 1: Fine-Tuning on Medical Data

PNEUMONIA class:

- Very high true positive rate, with the majority of pneumonia cases correctly identified.
- Very few false negatives, indicating that the model rarely misses pneumonia cases.

NORMAL class:

- A noticeable number of normal images are misclassified as pneumonia.
- This results in lower recall for the NORMAL class.



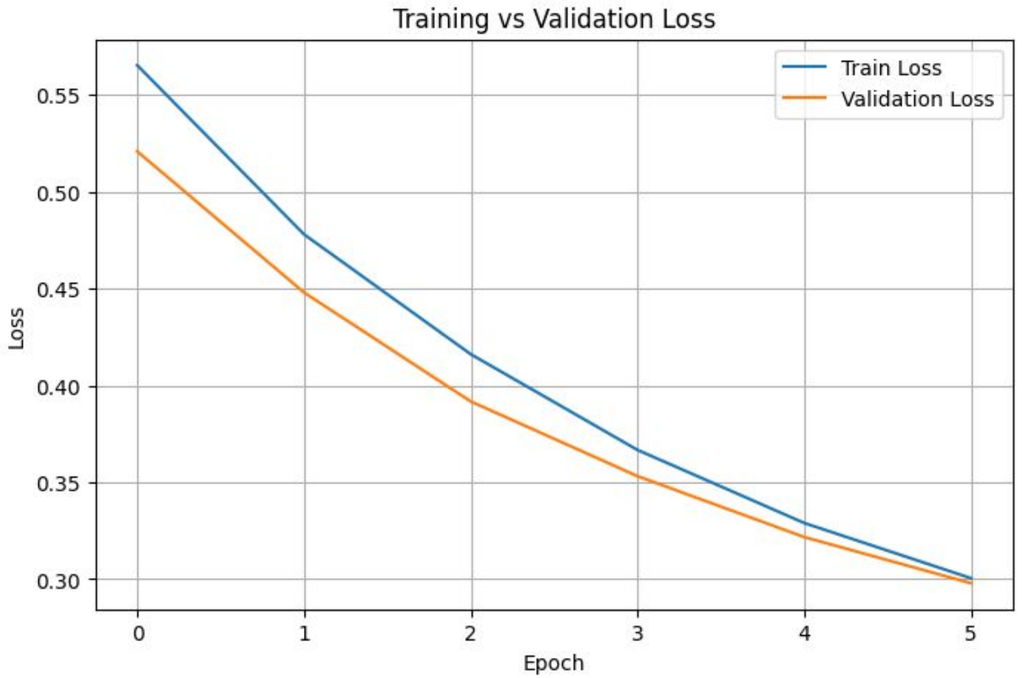
Experiment 2: Linear Evaluation on Medical Data

- Measure how well SimCLR representations generalize to medical imaging without adaptation.
- Entire backbone frozen
- Only linear classifier trained
- Class-weighted loss used

Experiment 2: Linear Evaluation on Medical Data

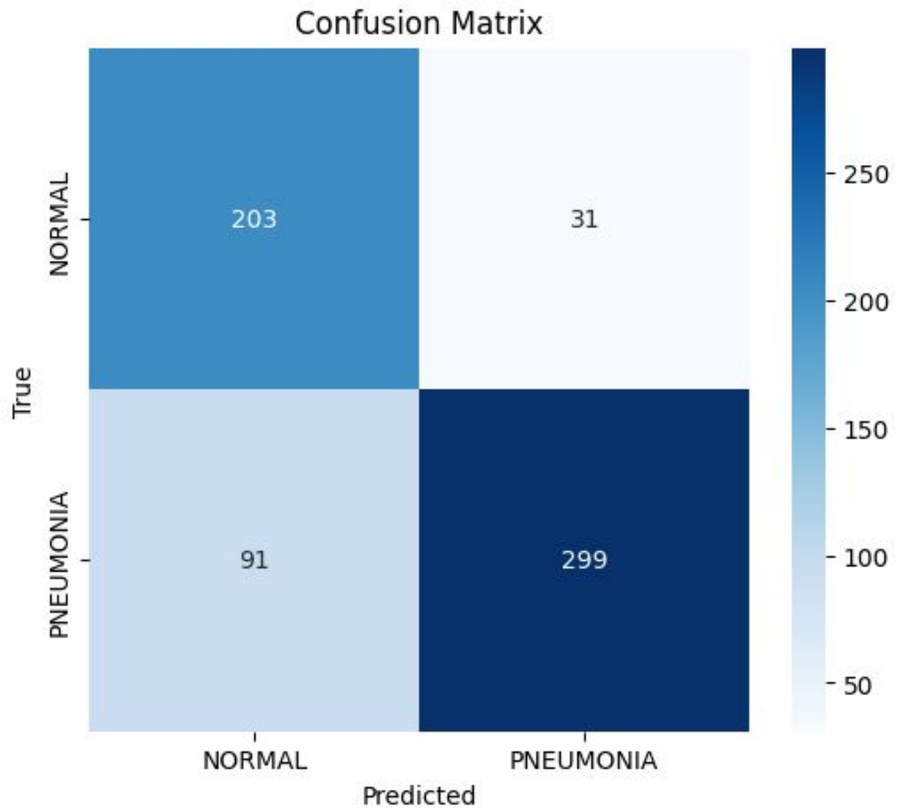
Final Test Performance

- Test Accuracy: 80.45%
- Convergence: The two curves are converging toward the same value by epoch 6.
- No Overfitting: There is no sign of overfitting here. Usually, overfitting is indicated by the Validation Loss starting to increase while the Train Loss continues to decrease.
- Generalization: Since the Validation Loss is very close to the Train Loss, the model is generalizing well to unseen data.



Experiment 2: Linear Evaluation on Medical Data

- The confusion matrix for linear evaluation on the medical dataset shows a more balanced error distribution compared to fine-tuning:
- NORMAL class:
  - Improved recall compared to Experiment 1.
  - Fewer normal images incorrectly classified as pneumonia.
- PNEUMONIA class:
  - Slightly reduced recall compared to fine-tuning.
  - More false negatives than Experiment 1, though still relatively low.



Experiment 3: Linear Evaluation on Non-Medical Data

- Assess representation quality when downstream data domain matches natural image statistics.
- Identical linear evaluation protocol
- No class imbalance correction required

Experiment 3: Linear Evaluation on Non-Medical Data

Final Test Performance

- Test Accuracy: 80.93%

Classification Report

- Balanced precision and recall across classes
- No significant bias observed

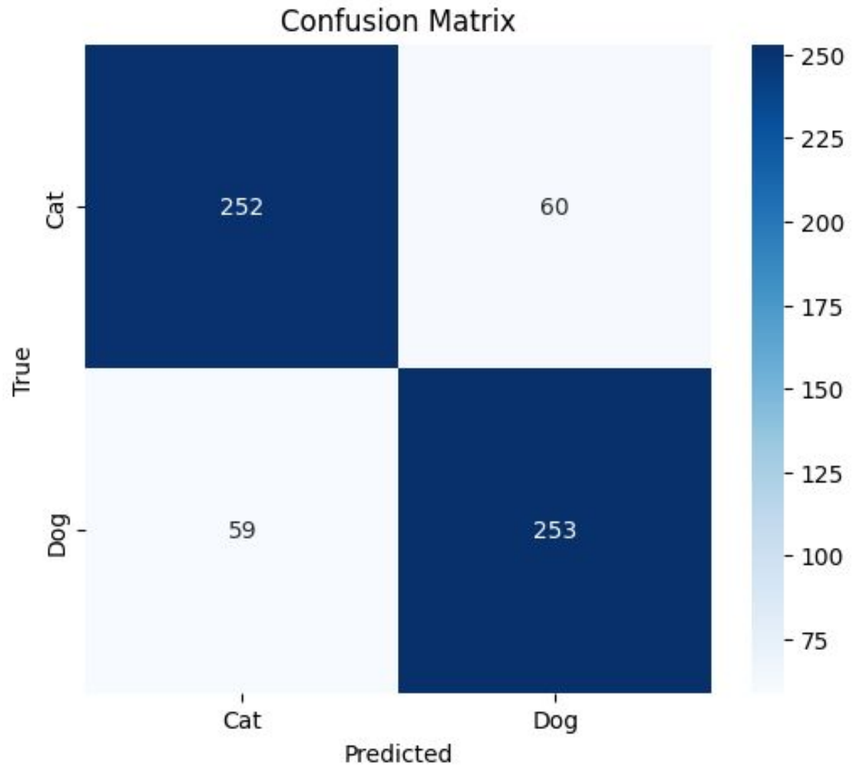


Experiment 3: Linear Evaluation on Non-Medical Data

Confusion Matrix Analysis

The confusion matrix for the non-medical (Cats vs Dogs) linear evaluation experiment shows:

- Nearly symmetric misclassification rates between cats and dogs.
- No dominant bias toward either class.
- Errors are evenly distributed, suggesting well-separated feature representations.



Related Works

- Several recent studies support the findings observed in this project
  - Research has shown that SimCLR can outperform supervised pre-training on medical image datasets when labeled data is limited (Azizi et al., 2021, Reed et al, 2022)
  - simclr - 94.92 % and supervised baseline - 93.68 % on chexpert dataset
  - simclr - 80.61 % and supervised baseline - 80.49 on BACH (histopathology)
- Several recent variants of SimCLR have been proposed specifically for grayscale medical images:
  - MedCLR (Ren et al., 2023)
    - Adapts SimCLR for medical imaging by replacing color-based augmentations with domain-appropriate transformations, such as rotations, intensity shifts, and noise injection.
    - Demonstrated improved performance on chest X-rays, CT scans, and MRI datasets compared to standard SimCLR.
    - 98.9 % accuracy on LC25000 (medical tissue) and 94.1 % accuracy on BCCD (blood cell images)

Conclusion

- No significant performance difference was observed between using medical and non-medical datasets when the data size was controlled.
- SimCLR representations generalized well across domains, indicating strong transferability of self-supervised features.
- Slight performance improvement was achieved with fine-tuning compared to linear evaluation.
- Domain-specific adaptations (e.g., MedCLR) further enhance performance but are not strictly necessary to achieve reasonable accuracy, as the experiments confirm.

Thank you